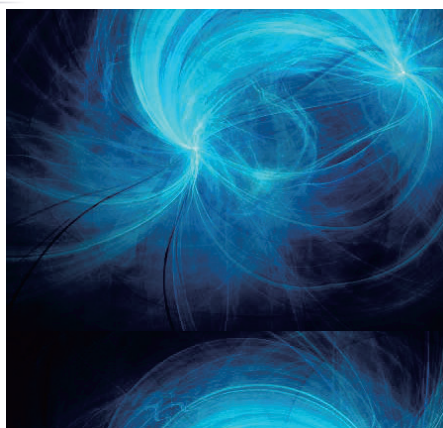


Unit 3: Electric Circuits

AP Physics C Electricity and Magnetism



Contents

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01 Current, Resistance, Power

05 Capacitors in Circuits

02 Electromotive Force and Internal Resistance

03 Ammeters and Voltmeters

04 Kirchhoff's Rules

TESTDAILY

Current, Resistance, Power



TESTDAILY

Current, Resistance, Power

3.1.1 — Current

3.1.4 — Energy and Power

3.1.2 — Resistance

3.1.3 — Resistors in Series and Parallel

TESTDAILY

3.1 Current, Resistance, Power

3.1.1 Current

3.1.1

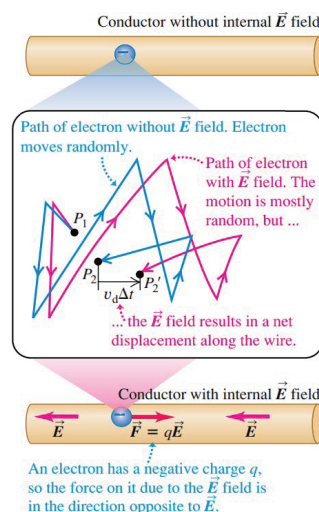
Definition

Current

- A current is any motion of charge from one region to another.
- In electrostatic situation the electric field is zero, and there is no current.
- Electrons move randomly in all directions of the order of 10^6 m/s.

3.1.1

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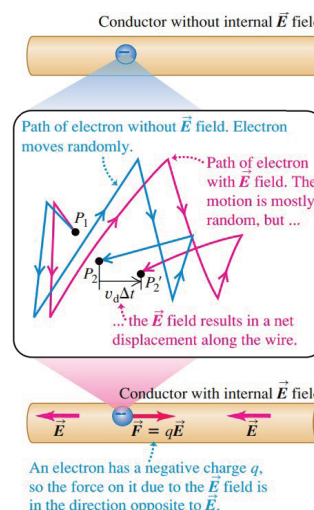
Definition

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Current

- If a constant steady electric field is established inside a conductor, there is a very slow net motion or **drift** of the moving charged particles as a group in the direction of the electric force
- This motion is described in terms of the drift velocity $\vec{v}_d$ of the particles. As a result, there is a net current in the conductor.
- v_d is often on the order of 10^{-4} m/s.

3.1.1

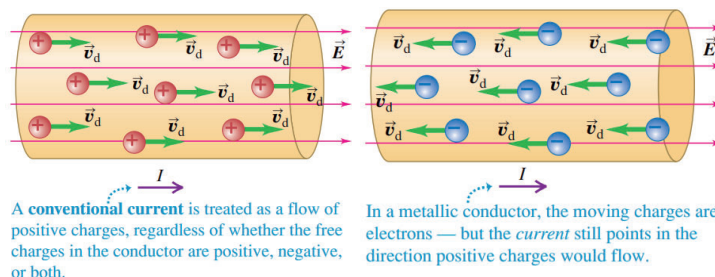


Property

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Current

- In different current-carrying materials, the charges of the moving particles may be positive or negative. We define the current to be in the direction in which there is a flow of **positive** charge.
- This choice or convention for the direction of current flow is called conventional current.



3.1.1

Definition

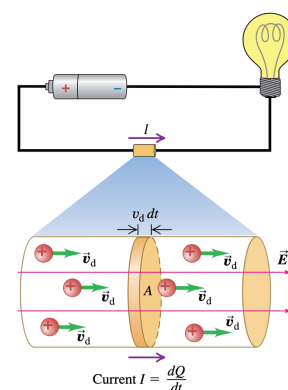
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Current (CED p63)

Physical quantity	Current 电流
Definition	The net charge flowing through the area per unit time
Symbol	I
Equation	$I = \frac{dQ}{dt}$
Unit	ampere, A 安培 $1 \text{ A} = 1 \text{ C/s}$

Caution: current is a scalar, not a vector.

3.1.1



Concept

TESTDAILY

Microscopic Definition of Current

- Let $n = \frac{N}{V}$ be the number of charge carriers per unit volume.
- The charge dQ that flows out of the end of the cylinder during time dt is

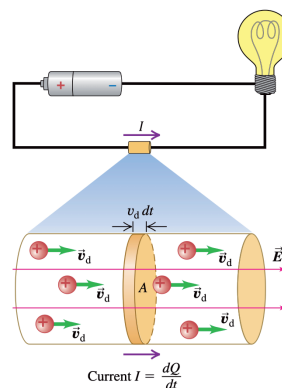
$$dQ = Nq = nVq = nAv_d dtq = (nqAv_d)dt$$

- And the current is

$$I = \frac{dQ}{dt} = nqAv_d$$

- For metal conductor

$$I = nev_d A$$



3.1.1

3.1 Current, Resistance, Power

3.1.2 Resistance

3.1.2

Property

TESTDAILY

Ohm's Law (CED p63) and Resistance (Young 13e p825)

- For some materials, especially metals, at a given temperature, current is proportional to the potential difference

$$I \propto \Delta V$$

- Ohm's Law

$$I = \frac{\Delta V}{R}$$

- A circuit device made to have a specific value of resistance between its ends is called a **resistor**.

Physical quantity	Resistance 电阻
Definition	The ratio of potential difference to current
Symbol	R
Equation	$R = \frac{\Delta V}{I}$
Unit	ohm, Ω 欧姆 $1 \Omega = 1 \text{ V/A}$

3.1.2

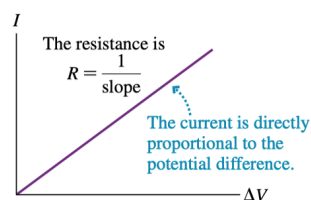
Property

TESTDAILY

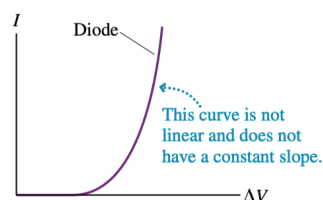
Resistance

A resistor that obeys Ohm's law is called an **ohmic** or **linear** resistor. The resistance is a constant. The graph of current as a function of potential difference (voltage) is a straight line. The slope of the line is $1/R$.

(a) Ohmic materials



(b) Nonohmic materials



3.1.2

Definition

TESTDAILY

Resistance and Resistivity

- The resistance is determined by the material of the conductor, and its geometry.

$$R = \rho \frac{L}{A}$$

where ρ is defined as the resistivity of the material.

3.1.2

Definition

TESTDAILY

Current Density

Physical quantity	Current density 电流密度
Definition	The current per unit cross-sectional area
Symbol	J
Equation	$J = \frac{I}{A}$
Unit	ampere per square meter, A/m^2

- The microscopic definition of current density

$$J = \frac{I}{A} = \frac{nqAv_d}{A} = nqv_d$$

- We can also define a vector current density $\vec{J}$ that includes the direction of the drift velocity

$$\vec{J} = nq\vec{v}_d$$

3.1.2

Concept

TESTDAILY

The Actual Ohm's Law

- The current density in a conductor depends on the electric field and on the properties of the material.
- For some materials, especially metals, at a given temperature, $\vec{j}$ is nearly directly proportional to $\vec{E}$ and the ratio of the magnitudes of $\vec{j}$ and $\vec{E}$ is constant. This relationship is called **Ohm's law**.

$$\vec{j} \propto \vec{E}$$

3.1.2

Definition

TESTDAILY

Resistivity (Young 13e p823)

Physical quantity	Resistivity 电阻率
Definition	The ratio of the magnitudes of electric field and current density
Symbol	ρ
Equation	$\rho = \frac{E}{j}$
Unit	ohm meter, $\Omega \cdot \text{m}$

- Derivation:

$$E = \rho j$$

$$\frac{\Delta V}{L} = \rho \frac{I}{A}$$

$$\frac{\Delta V}{I} = \rho \frac{L}{A}$$

$$R = \rho \frac{L}{A}$$

3.1.2

Property

TESTDAILY

Resistivity

Resistivities at Room Temperature (20°C)

Substance			ρ ($\Omega \cdot \text{m}$)	Substance		ρ ($\Omega \cdot \text{m}$)
Conductors				Semiconductors		
Metals	Silver		1.47×10^{-8}	Pure carbon (graphite)		3.5×10^{-5}
	Copper		1.72×10^{-8}	Pure germanium		0.60
	Gold		2.44×10^{-8}	Pure silicon		2300
	Aluminum		2.75×10^{-8}	Insulators		
	Tungsten		5.25×10^{-8}		Amber	5×10^{14}
	Steel		20×10^{-8}		Glass	$10^{10} - 10^{14}$
	Lead		22×10^{-8}		Lucite	$> 10^{13}$
	Mercury		95×10^{-8}	Mica	$10^{11} - 10^{15}$	
Alloys	Manganin (Cu 84%, Mn 12%, Ni 4%)		44×10^{-8}	Quartz (fused)		75×10^{16}
	Constantan (Cu 60%, Ni 40%)		49×10^{-8}	Sulfur		10^{15}
	Nichrome		100×10^{-8}	Teflon		$> 10^{13}$
			Wood		$10^8 - 10^{11}$	

3.1.2

Exercise

TESTDAILY

Comparison of Resistors

- Different dimensions:

A cylindrical rod has resistance R . If we triple its length and diameter, what is its resistance, in terms of R ?

- Different materials:

Resistor 1 has length L and diameter D . Resistor 2 has length $L/2$ and diameter $D/2$. If they have the same resistance, what is the ratio of resistivities ρ_1/ρ_2 ?



3.1 Current, Resistance, Power

3.1.3 Resistors in Series and Parallel



Introduction

TESTDAILY

Resistors in Series and Parallel

- For any combination of resistors we can always find a single resistor that could replace the combination and result in the same total current and potential difference. The resistance of this single resistor is called the **equivalent resistance** of the combination.

$$R_{eq} = \frac{\Delta V}{I}$$



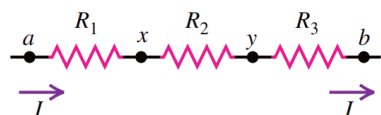
Concept

TESTDAILY

Resistors in Series and Parallel

In series: several circuit elements such as resistors, batteries, and motors are connected in sequence, with only **a single current path** between the points.

(a) R_1 , R_2 , and R_3 in series



$$I = I_1 = I_2 = I_3$$

$$\Delta V = \Delta V_1 + \Delta V_2 + \Delta V_3$$

$$R = R_1 + R_2 + R_3$$



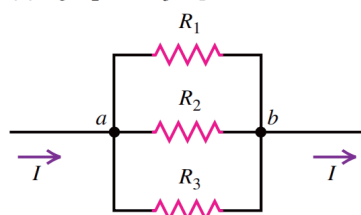
Concept

TESTDAILY

Resistors in Series and Parallel

In parallel: each resistor provides an alternative path between the points.

(b) R_1 , R_2 , and R_3 in parallel



$$\Delta V = \Delta V_1 = \Delta V_2 = \Delta V_3$$

$$I = I_1 + I_2 + I_3$$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$



Property

TESTDAILY

Resistors in Series and Parallel

- For resistors in series, the equivalent resistance is the sum of the resistance of all resistors.

$$R_s = \sum_i R_i$$

$$R_s > R_i$$

- For resistors in parallel, the reciprocal of equivalent resistance is the sum of the reciprocal of resistance of all resistors.

$$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$$

$$R_p < R_i$$



Property

TESTDAILY

Resistors in Series and Parallel

- For two resistors

$$R_s = R_1 + R_2$$

$$R_p = \frac{R_1 R_2}{R_1 + R_2}$$

- For identical resistors

$$R_s = NR$$

$$R_p = \frac{R}{N}$$



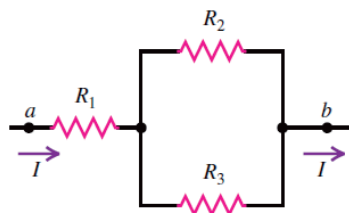
Method

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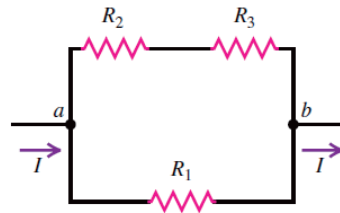
Resistors in Series and Parallel

“Collapse” series and parallel resistors

(c) R_1 in series with parallel combination of R_2 and R_3



(d) R_1 in parallel with series combination of R_2 and R_3

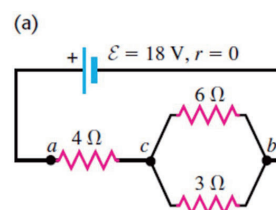


Example

TESTDAILY

Young_Chapter26_Example26.1

Find the equivalent resistance of the network and the current in each resistor.



Answer: $6\ \Omega$, $3\ \text{A}$, $2\ \text{A}$, $1\ \text{A}$.



Property

TESTDAILY

Resistors in Series and Parallel

- For resistors in series, the current is constant, so the voltage across each resistor is proportional to the resistance.

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

- For resistors in parallel, the voltage across each resistor is equivalent, so the current carried by resistors is inversely proportional to their resistances.

$$\frac{I_1}{I_2} = \frac{R_2}{R_1}$$



3.1 Current, Resistance, Power

3.1.4 Energy and Power



Concept

TESTDAILY

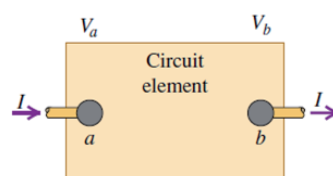
Energy and Power

- As an amount of charge dq passes through the circuit element, there is a change in potential energy equal to $dq\Delta V$
- Power is the **rate** at which energy is delivered to a circuit element.

$$dU = dq\Delta V = (Idt)\Delta V$$

$$P = \frac{dU}{dt} = I\Delta V$$

SI unit: watt, W.



Energy and Power

- If the circuit element is a resistor, the potential difference is $\Delta V = IR$, the electric power delivered to the resistor by the circuit is

$$P = I\Delta V = I^2R = \frac{\Delta V^2}{R}$$

- Energy dissipated over time

$$E = \int P dt = P_{avg} \Delta t$$

- The moving charges collide with atoms in the resistor and transfer energy to these atoms, **increasing the internal energy of the material**. Either the temperature of the resistor increases or there is a flow of heat out of it, or both.



Example

Young_Chapter26_Example26.2

Two identical light bulbs, each with resistance $R = 2 \Omega$, are connected to a voltage $\Delta V = 8 \text{ V}$. Find the current through each bulb, the potential difference across each bulb, and the power delivered to each bulb and to the entire network if the bulbs are connected (a) in series and (b) in parallel. (c) Suppose one of the bulbs burns out; that is, its filament breaks and current can no longer flow through it. What happens to the other bulb in the series case? In the parallel case?

Answer: 2A, 4 V, 8 W; 4 A, 8 V, 32 W.

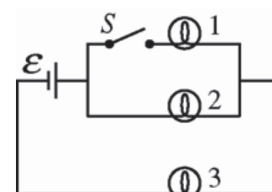


Exercise

2008_Intl_MCQ_13

The three lightbulbs in the circuit above are identical, and the battery has zero internal resistance. When switch S is closed to cause bulb 1 to light, which of the other two bulbs increase(s) in brightness?

- (A) Neither bulb
- (B) Bulb 2 only
- (C) Bulb 3 only
- (D) Both bulbs
- (E) It cannot be determined without knowing the emf of the battery.



Electromotive Force and Internal Resistance

3.2

Concept

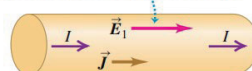
Electromotive Force

For a conductor to have a steady current, it must be part of a path that forms a closed loop or complete circuit.

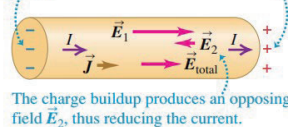
If an electric field is produced inside a conductor that is *not* part of a complete circuit, current flows for only a very short time.

3.2

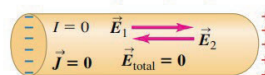
(a) An electric field $\vec{E}_1$ produced inside an isolated conductor causes a current.



(b) The current causes charge to build up at the ends.



(c) After a very short time $\vec{E}_2$ has the same magnitude as $\vec{E}_1$; then the total field is $\vec{E}_{\text{total}} = 0$ and the current stops completely.



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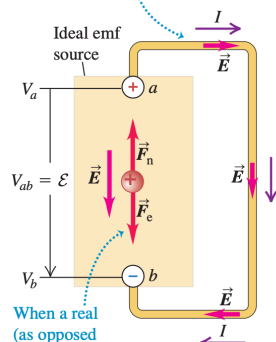
Concept

Electromotive Force

- If a charge goes around a complete circuit and returns to its starting point, the **potential energy must be the same** at the end of the round trip as at the beginning.
- So there must be some part of the circuit in which the potential energy **increases**.
- The influence that makes current flow from lower to higher potential is called **electromotive force**. (abbreviated EMF or emf). Every complete circuit with a steady current must include some device that provides emf. Such a device is called a **source of emf**.

3.2

Potential across terminals creates electric field in circuit, causing charges to move.



When a real (as opposed to ideal) emf source is connected to a circuit, V_{ab} and thus F_e fall, so that $F_n > F_e$ and $\vec{F}_n$ does work on the charges.

TESTDAILY

Definition

TESTDAILY

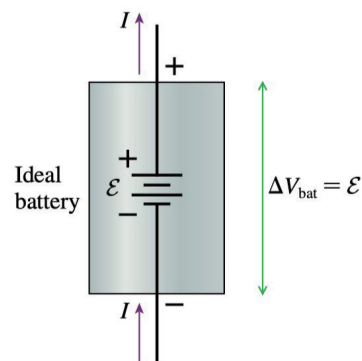
Electromotive Force (Young 13e p829)

Physical quantity	Electromotive force 电动势
Definition	The work done to move a unit charge across the terminals of the battery.
Symbol	ε
Equation	$\varepsilon = \frac{W}{q}$
Unit	volt, V 伏特 $1 \text{ V} = 1 \text{ J/C}$

- Ideally, there is no energy dissipated within the battery.

$$W = \Delta U$$

$$\varepsilon = \Delta V$$



3.2

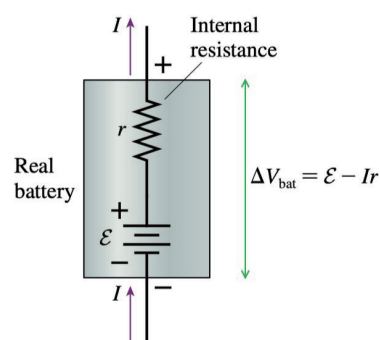
Concept

TESTDAILY

Real Source of Emf and Internal Resistance

- Real sources of emf in a circuit don't behave in exactly the way we have described; the potential difference across a real source in a circuit is not equal to the emf.
- A real battery is modelled as an ideal source and emf and an internal resistance in series.
- Terminal voltage is smaller than the emf.

$$\Delta V = \varepsilon - Ir$$



3.2

Ammeters and Voltmeters

TESTDAILY

3.3

Ammeters and Voltmeters

3.3.1 — Real Ammeters and Voltmeters

3.3.2 — Design Ammeters and Voltmeters

TESTDAILY

3.3 Ammeters and Voltmeters

3.3.1 Real Ammeters and Voltmeters



Concept

TESTDAILY

Ammeters and Voltmeters

- Ammeter
 - A current-measuring instrument.
 - Placed in series within the branch to be measured.
 - Ideal: negligible (zero) resistance.
- Voltmeter
 - A voltage-measuring device.
 - Placed in a parallel arrangement with the circuit element being measured.
 - Ideal: very large (infinite) resistance.

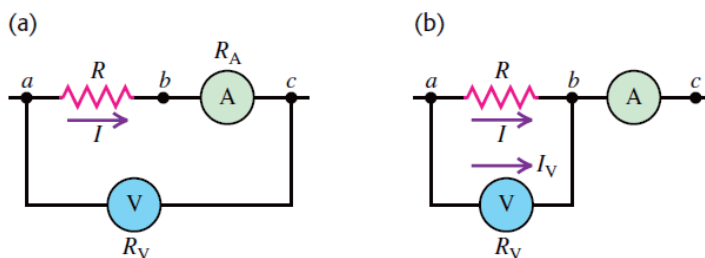


Concept

TESTDAILY

Real Ammeters and Voltmeters

- Real ammeter: small but **nonzero** resistance.
- Real voltmeter: large but **finite** resistance.
- Assess qualitatively the effect of real meters on a circuit.



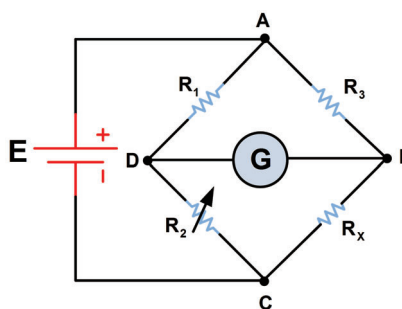
3.3.1

Concept

TESTDAILY

Wheatstone Bridge

- Purpose: calculate unknown resistance.
- When no current flows through the **galvanometer**, the ratio of the resistances are equal.



3.3.1

3.3 Ammeters and Voltmeters

3.3.2 Design Ammeters and Voltmeters

3.3.2

TESTDAILY

Example**Young_Chapter26_Example26.8**

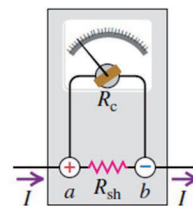
What shunt resistance is required to make the 1.00-mA, 20.0- Ω meter described above into an ammeter with a range of 0 to 50.0 mA?

Answer: 0.408 Ω

3.3.2

TESTDAILY

(a) Moving-coil ammeter

**Example****Young_Chapter26_Example26.9**

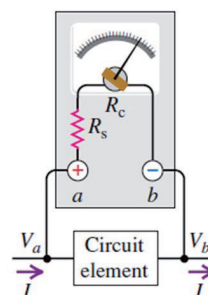
What series resistance is required to make the 1.00-mA, 20.0- Ω meter described above into a voltmeter with a range of 0 to 10.0 V?

Answer: 9980 Ω

3.3.2

TESTDAILY

(b) Moving-coil voltmeter

**Kirchhoff's Rules**

3.4

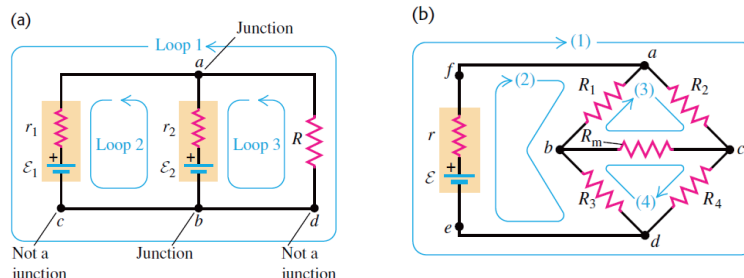
TESTDAILY

Introduction

TESTDAILY

Kirchhoff's Rules

- To compute the currents in complicated networks that cannot be reduced using conventional methods, we'll use the techniques developed by the German physicist Gustav Robert Kirchhoff.
- A junction in a circuit is a point where three or more conductors meet. A loop is any closed conducting path.



3.4

Property

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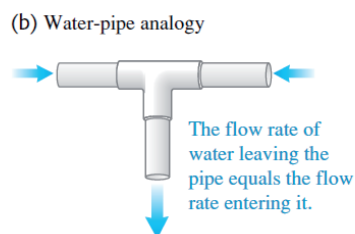
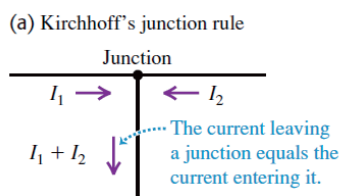
The Junction Rule

- Kirchhoff's junction/**current** rule: The algebraic sum of the currents into any junction is zero.

$$\sum I = 0$$

$$\sum I_{in} = \sum I_{out}$$

- The junction rule is based on conservation of electric charge.



3.4

Property

TESTDAILY

The Loop Rule

- Kirchhoff's loop/**voltage** rule: The algebraic sum of the potential differences in any loop, including those associated with emfs and those of resistive elements, must equal zero.

$$\sum \Delta V = 0$$

- This must hold for the potential at a point to have a definite value.
- The junction rule is based on conservation of energy. The net change of potential energy for a charge q making a round trip around a complete circuit must be zero. Hence the net change in potential around the circuit must be zero.

3.4

Property

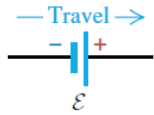
TESTDAILY

The Loop Rule

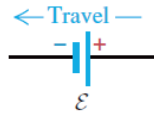
Traveling around the loop can be in arbitrary direction, clockwise or counterclockwise.

(a) Sign conventions for emfs

$+\mathcal{E}$: Travel direction
from $-$ to $+$:

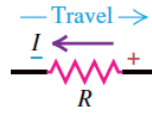


$-\mathcal{E}$: Travel direction
from $+$ to $-$:

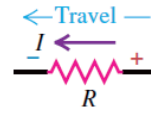


(b) Sign conventions for resistors

$+IR$: Travel *opposite*
to current direction:



$-IR$: Travel *in*
current direction:



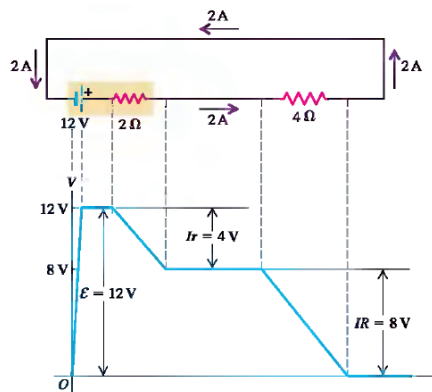
3.4

Theorem

TESTDAILY

Potential Changes around a Circuit

$$\mathcal{E} - Ir - IR = 0$$



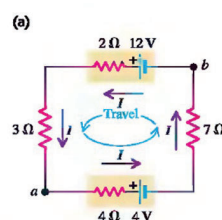
3.4

Example

TESTDAILY

Young_Chapter26_Example26.3

The circuit shown in figure contains two batteries, each with an emf and an internal resistance, and two resistors. Find (a) the current in the circuit, (b) the potential difference V_{ab} ,



Answer: 0.5 A, 9.5 V

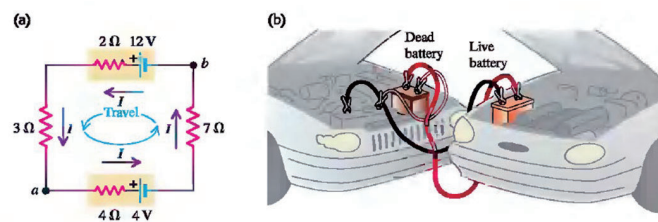
3.4

Example

TESTDAILY

Young_Chapter26_Example26.3

and (c) the power output of the emf of each battery.



Answer: 6 W, -2 W.

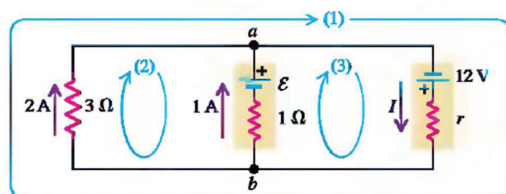
3.4

Example

TESTDAILY

Young_Chapter26_Example26.4

Find the unknown current I , the internal resistance r , and the emf $\mathcal{E}$.



Answer: 3 A, 2 Ω, -5 V

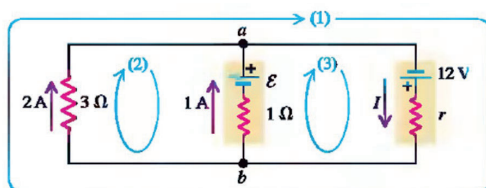
3.4

Example

TESTDAILY

Young_Chapter26_Example26.5

Find the power delivered by the 12-V power supply and by the battery being recharged, and find the power dissipated in each resistor.



Answer: 18 W, 6 W, 12 W.

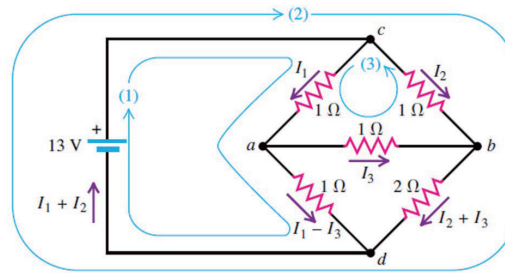
3.4

Example

TESTDAILY

Young_Chapter26_Example26.6

Figure below shows a “bridge” circuit. Find the current in each resistor and the equivalent resistance of the network of five resistors.



Answer: 6 A, 5 A, -1 A, 1.2 Ω

3.4

Capacitors in Circuits

3.5

Capacitors in Circuits

3.5.1 — Combinations of Capacitors

3.5.2 — RC Circuit: Charging a Capacitor

3.5.3 — RC Circuit: Discharging a Capacitor

TESTDAILY

3.5 Capacitors in Circuits

3.5.1 Combinations of Capacitors

3.5.1

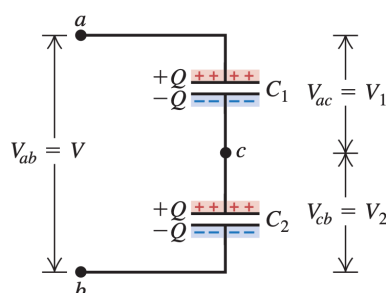
Concept

TESTDAILY

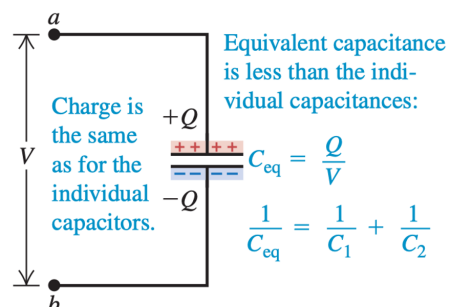
Combinations of Capacitors

Capacitors in series:

- The capacitors have the same charge Q .
- Their potential differences add:
 $V_{ac} + V_{cb} = V_{ab}$.



3.5.1



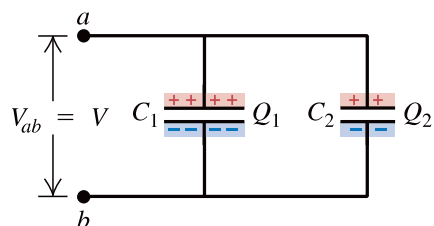
Concept

TESTDAILY

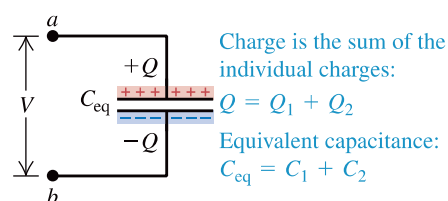
Combinations of Capacitors

Capacitors in parallel:

- The capacitors have the same potential V .
- The charge on each capacitor depends on its capacitance: $Q_1 = C_1 V$, $Q_2 = C_2 V$.



3.5.1



Property

TESTDAILY

Combinations of Capacitors

Capacitors in series

$$\frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots$$

Capacitors in parallel

$$C_p = C_1 + C_2 + C_3 + \dots$$

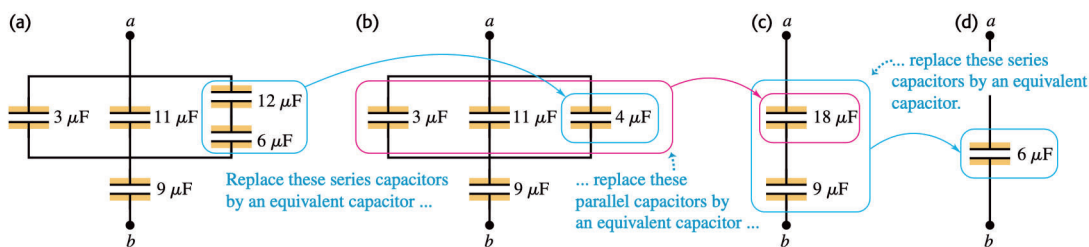


Example

TESTDAILY

Young_Chapter24_Example24.6

Find the equivalent capacitance of the five-capacitor network shown in the figure.

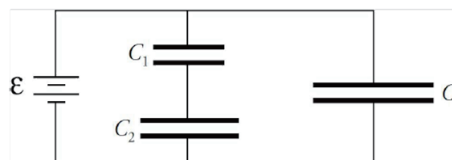
Answer: $6\mu\text{F}$ 

Example

TESTDAILY

Princeton_2020_Chapter13_Example21

Find the charge stored and the voltage across each capacitor in the following circuit, given that $\mathcal{E} = 180\text{ V}$, $C_1 = 30\text{ }\mu\text{F}$, $C_2 = 60\text{ }\mu\text{F}$, and $C_3 = 90\text{ }\mu\text{F}$.

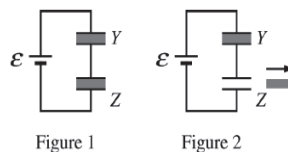
Answer: 3.6 mC , 120 V ; 3.6 mC , 60 V ; 16.2 mC , 180 V 

Exercise

TESTDAILY

2016_Intl_MCQ_33

Two identical capacitors, Y and Z, are connected in series with an ideal battery, as shown in Figure 1 above, and fully charged. Each capacitor has a dielectric slab of dielectric constant $\kappa > 1$ between its plates. If the dielectric slab is removed from capacitor Z, which of the following describes what happens to the voltage across each capacitor?



Voltage across Capacitor Y	Voltage across Capacitor Z
(A) Increases	Decreases
(B) Increases	Increases
(C) Remains the same	Remains the same
(D) Decreases	Decreases
(E) Decreases	Increases

3.5.1

3.5 Capacitors in Circuits

3.5.2 RC Circuit: Charging a Capacitor

3.5.2

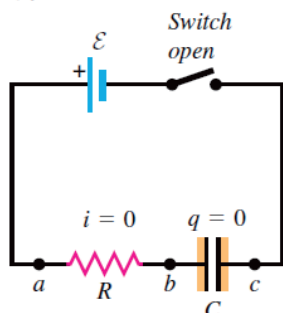
Concept

TESTDAILY

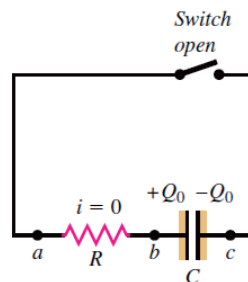
RC Circuit

- A circuit such as this that has a resistor and a capacitor in series is called an R-C circuit.

(a) Capacitor initially uncharged



(a) Capacitor initially charged



3.5.2

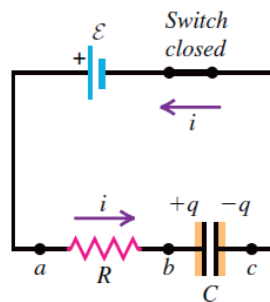
Concept

TESTDAILY

Charging a Capacitor

- The capacitor has no initial charge.

	$q(t)$	$ \Delta V_C $	$ \Delta V_R $	$i(t)$
$t = 0$				
$t > 0$				
$t = \infty$				



3.5.2

An uncharged capacitor acts like a “wire”.
A fully charged capacitor acts like an “open switch”.

Concept

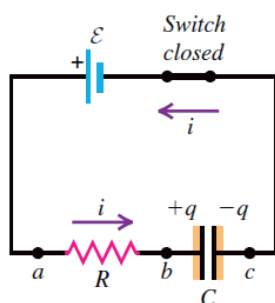
TESTDAILY

Charging a Capacitor

- Kirchhoff's loop rule gives

$$\mathcal{E} - iR - \frac{q}{C} = 0$$

- Solve a differential equation



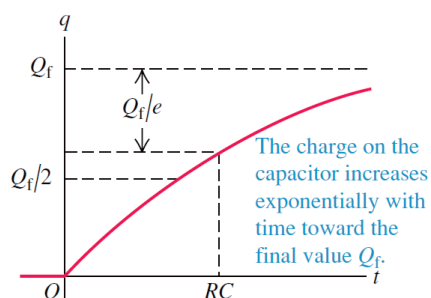
3.5.2

Property

TESTDAILY

Charging a Capacitor

$$q = C\mathcal{E}(1 - e^{-t/RC})$$



3.5.2

Property

TESTDAILY

Charging a Capacitor

- After a time equal to RC , the capacitor charge has reached $(1 - 1/e) = 0.632$ of its final value
 $Q_f = C\varepsilon$.
- The product RC is therefore a measure of how quickly the capacitor charges.
- Time constant τ is defined as

$$\tau = RC$$

$$q(t) = Q_f(1 - e^{-t/\tau})$$

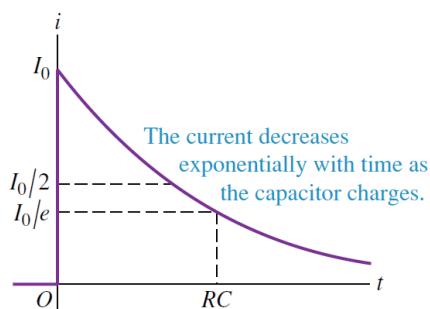


Property

TESTDAILY

Charging a Capacitor

$$i = \frac{dq}{dt} = \frac{\varepsilon}{R} e^{-t/RC} = I_0 e^{-t/\tau}$$



Example

TESTDAILY

Young_Chapter26_Example26.12

A $10\text{-M}\Omega$ resistor is connected in series with a $1.0\text{-}\mu\text{F}$ capacitor and a battery with emf 12.0 V . Before the switch is closed at time $t = 0$, the capacitor is uncharged.

- What is the time constant?
- What fraction of the final charge Q_f is on the capacitor at $t = 46\text{ s}$?
- What fraction of the initial current I_0 is still flowing at $t = 46\text{ s}$?

Answer: 10 s, 0.99, 0.01

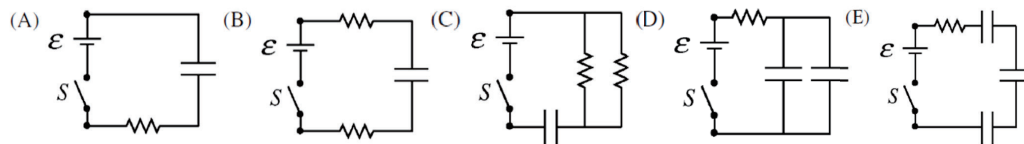


Example

TESTDAILY

2016_Intl_MCQ_15

Identical resistors and identical uncharged capacitors are connected to identical ideal batteries of emf $\mathcal{E}$ in the circuits shown below. Each circuit has a switch S in the open position. In which circuit will the capacitors reach half their maximum charge in the least amount of time when switch S is closed?



3.5.2

3.5 Capacitors in Circuits

3.5.3 RC Circuit: Discharging a Capacitor

3.5.3

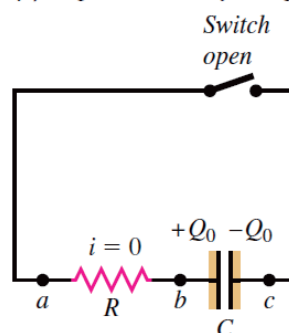
Property

TESTDAILY

Discharging a Capacitor

- The capacitor has acquired a charge, i.e. $t = 0$, $q = Q_0$

	$q(t)$	$ \Delta V_C $	$ \Delta V_R $	$i(t)$
$t = 0$				
$t > 0$				
$t = \infty$				



An initially charged capacitor acts like a “battery”.
A discharged capacitor acts like a “wire”.

3.5.3

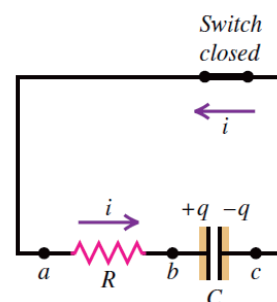
Property

TESTDAILY

Discharging a Capacitor

- Kirchhoff's loop rule gives us

$$-iR - \frac{q}{C} = 0$$



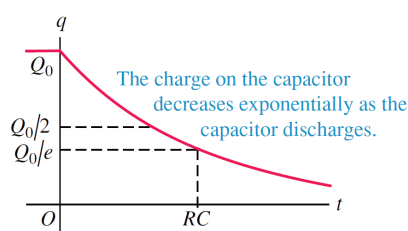
3.5.3

Property

TESTDAILY

Discharging a Capacitor

$$q = Q_0 e^{-t/RC} = Q_0 e^{-t/\tau}$$



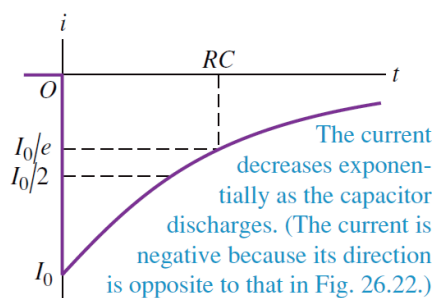
3.5.3

Property

TESTDAILY

Discharging a Capacitor

$$i = \frac{dq}{dt} = -\frac{Q_0}{RC} e^{-t/RC} = I_0 e^{-t/\tau}$$



3.5.3

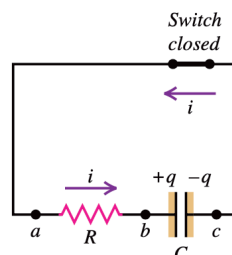
Example

TESTDAILY

Young_Chapter26_Example26.13

The resistor and capacitor of Example 26.12 are reconnected as shown. The capacitor has an initial charge of $5.0 \mu\text{C}$ and is discharged by closing the switch at $t = 0$.

- (a) At what time will the charge be equal to $0.50 \mu\text{C}$?
 (b) What is the current at this time?



Answer: 23 s, $-5 \times 10^{-8} \text{ A}$

3.5.3

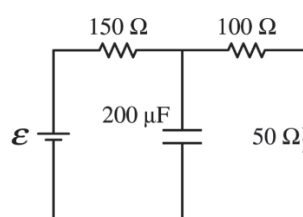
Example

TESTDAILY

2019_NA_Set1_FRQ_2 (d)i

The battery has voltage $\mathcal{E}$ of 12 V, and the circuit is allowed to reach **steady** state. Calculate the current through the 50Ω resistor.

A fully charged capacitor acts like an “open switch”.



Answer: 0.04 A

3.5.3

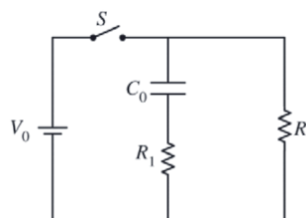
Example

TESTDAILY

2017_NA_FRQ_2

In the circuit above, an ideal battery of voltage V_0 is connected to a capacitor with capacitance C_0 and resistors with resistances R_1 and R_2 , with $R_1 > R_2$. The switch S is open, and the capacitor is initially uncharged.

- (a) The switch is closed at time $t = 0$. On the axes below, sketch the charge q on the capacitor as a function of time t . Explicitly label any intercepts, asymptotes, maxima, or minima with numerical values or algebraic expressions, as appropriate.



3.5.3

Example

TESTDAILY

2017_NA_FRQ_2

(b) On the axes below, sketch the current I through each resistor as a function of time t . Clearly label the two curves as I_1 and I_2 , the currents through resistors R_1 and R_2 , respectively. Explicitly label any intercepts, asymptotes, maxima, or minima with numerical values or algebraic expressions, as appropriate.

An uncharged capacitor acts like a “wire”.



3.5.3

Example

TESTDAILY

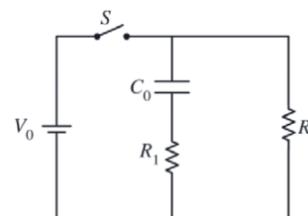
2017_NA_FRQ_2

The circuit is constructed using an ideal 1.5 V battery, an $80\ \mu\text{F}$ capacitor, and resistors $R_1 = 150\ \Omega$ and $R_2 = 100\ \Omega$. The switch is closed, allowing the capacitor to fully charge. The switch is then opened, allowing the capacitor to discharge.

(c) The time it takes to charge the capacitor to 50% of its maximum charge is Δt_C . The time it takes for the capacitor to discharge to 50% of its maximum charge is Δt_D . Which of the following correctly relates the two time intervals?

_____ $\Delta t_C > \Delta t_D$ _____ $\Delta t_C = \Delta t_D$ _____ $\Delta t_C < \Delta t_D$

Justify your answer.



3.5.3

Example

TESTDAILY

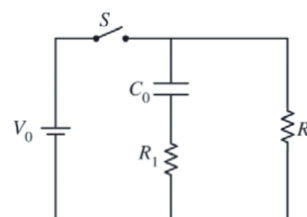
2017_NA_FRQ_2

(d)

- Calculate the current through resistor R_2 immediately after the switch is opened.
- Is the current through resistor R_2 increasing, decreasing, or constant immediately after the switch is opened?

_____ Increasing _____ Decreasing _____ Constant

Justify your answer.



3.5.3

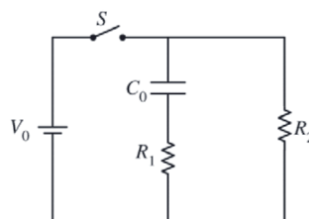
Example

TESTDAILY

2017_NA_FRQ_2

(e)

- Calculate the energy stored in the capacitor immediately after the switch is opened.
- Calculate the energy dissipated by resistor R_1 as the capacitor completely discharges.



3.5.3

Answer to MCQ

TESTDAILY

Section	Source	Answer
3.1.4	2008_Intl_MCQ_13	C
3.5.1	2016_Intl_MCQ_33	E
3.5.2	2016_Intl_MCQ_15	E

3

Answer to FRQ

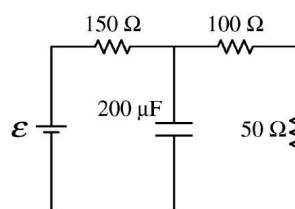
TESTDAILY

2019_NA_Set1_FRQ_2 (d)i

(d)

- LO CNV-7.B.a, SP 6.A, 6.C
2 points

The $200\ \Omega$ resistor in the circuit in Figure 2 is replaced with a $200\ \mu\text{F}$ capacitor, as shown on the right, and the circuit is allowed to reach steady state. Calculate the current through the $50\ \Omega$ resistor.



For substituting the voltage consistent with part (c) into Ohm's law	1 point
For correctly calculating the equivalent resistance of the circuit	1 point
$I = \frac{\mathcal{E}}{R_{tot}} = \frac{(12.1\ \text{V})}{(150\ \Omega + 100\ \Omega + 50\ \Omega)} = 40.3\ \text{mA}$	

3

Answer to FRQ

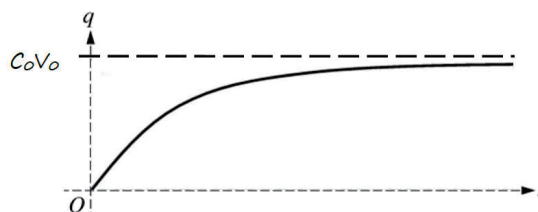
TESTDAILY

2017_NA_FRQ_2

15 points total

Distribution
of points

(a) 3 points



For a concave down curve starting at the origin
For a horizontal asymptote
For properly labeling the horizontal asymptote

1 point
1 point
1 point

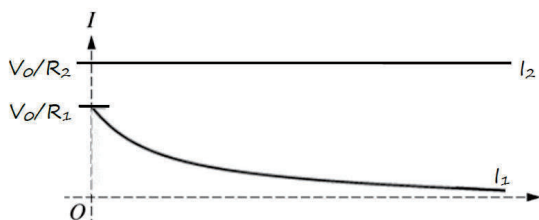
3

Answer to FRQ

TESTDAILY

2017_NA_FRQ_2

(b) 4 points



For showing that I_2 is horizontal with the correct vertical intercept
For showing that I_1 is concave up and asymptotic to $I = 0$
For correctly labeling the vertical intercept of the I_1 graph
For drawing the I_2 graph always above the I_1 graph

1 point
1 point
1 point
1 point

3

Answer to FRQ

TESTDAILY

2017_NA_FRQ_2

(c) 2 points

Select " $\Delta t_C < \Delta t_D$ "

For indicating that the equivalent resistance during discharging is greater than during charging

1 point

For a statement relating the greater resistance to the greater time constant, or to a smaller current

1 point

Example: Because the resistance as the capacitor discharges is greater than when it charges, the time constant is larger for discharging. Therefore, the time to charge to 50% of its maximum charge is less than the time to discharge to 50% of its maximum charge.

3

Answer to FRQ

TESTDAILY

2017_NA_FRQ_2

- (d)
i. 2 points

For using a correct statement of Ohm's law or loop rule with the switch open

$$V = IR$$

$$I = \frac{V_C}{R_{tot}} = \frac{V_0}{R_1 + R_2}$$

Substitute correct values into equation above

$$I = \frac{(1.5 \text{ V})}{(100 \, \Omega + 150 \, \Omega)}$$

For a correct answer

$$I = 6.0 \text{ mA with units, or } 0.006 \text{ without units}$$

1 point

1 point

3

Answer to FRQ

TESTDAILY

2017_NA_FRQ_2

- ii. 1 point

For a correct justification

Example: Once the capacitor begins to discharge, the charge stored in the capacitor will decrease and the potential difference across the capacitor decreases. Therefore, the current through resistor R_2 decreases.

1 point

3

Answer to FRQ

TESTDAILY

2017_NA_FRQ_2

- (e)
i. 1 point

For correctly substituting into an equation for the energy stored in a capacitor, including units for numerical values

$$U = \frac{1}{2}CV^2 = \frac{1}{2}(80 \, \mu\text{F})(1.5 \text{ V})^2 = 90 \, \mu\text{J}$$

1 point

- ii. 2 points

For setting up a valid equation or argument to calculate the energy dissipated by resistor R_1

$$E_{dis} = Pt = I^2 Rt \therefore E_{dis} \sim R \therefore E_{dis,1} = E_{dis,tot} \frac{R_1}{(R_1 + R_2)}$$

For correctly substituting into the equation above

$$E_{dis,1} = (90 \, \mu\text{J}) \frac{(150 \, \Omega)}{(150 \, \Omega + 100 \, \Omega)} = 54 \, \mu\text{J}$$

1 point

1 point

3

Alternate Solution on next page.

Answer to FRQ

TESTDAILY

2017_NA_FRQ_2

(e)(ii) continued

Alternate Solution

Alternate Points

Derive an equation for the current through resistor R_1 as a function of time

$$I(t) = I_{\text{MAX}} e^{-t/\tau} = \frac{V_{\text{MAX}}}{(R_1 + R_2)} e^{-t/(R_1 + R_2)C} = \frac{(1.5 \text{ V})}{(250 \, \Omega)} e^{-t/(250 \, \Omega)(80 \, \mu\text{F})}$$

$$I(t) = (6.0 \text{ mA}) e^{-t/(0.02)}$$

For using a correct formula with integral calculus to calculate the energy dissipated by resistor R_1

1 point

$$E = \int P dt = \int I^2 R dt = \int I(t)^2 R_1 dt = \int ((6.0 \text{ mA}) e^{-t/(0.02)})^2 (150 \, \Omega) dt$$

For integrating the above equation with correct limits or constants of integration

1 point

$$E = (0.0054) \int_{t=0}^{t=\infty} (e^{-t/(0.02)})^2 dt = (0.0054) \int_{t=0}^{t=\infty} e^{-100t} dt = (0.0054) \left[\frac{e^{-100t}}{(-100)} \right]_{t=0}^{t=\infty}$$

$$E = (-5.4 \times 10^{-5}) (e^{-\infty} - e^0) = 54 \, \mu\text{J}$$

3

TESTDAILY

3

TESTDAILY

3